MILITARY SPECIFICATION SHEET

PLASTIC SHEET, LAMINATED, METAL-CLAD (FOR PRINTED WIRING BOARDS), BASE MATERIAL GY, GLASS BASE WOVEN, POLYTETRAFLUOROETHYLENE RESIN, FLAME RESISTANT, FOR MICROWAVE APPLICATION, COPPER-CLAD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the plastic sheets described herein shall consist of this specification and the latest issue of MIL-P-13949.

REQUIREMENTS:

Property values: See table I.

TABLE I. Property values.

Property to be tested	Test condition 1/	Core thickness	
		Less than 0.030 in. to 0.010 in.	0.030 in. and over
1. Peel strength, minimum in lbs-per-inch width			
A. 1/2-oz/ft^2 copper			
As received	A	4.0	4.0
After thermal stress	A	4.0	4.0
At elevated temperature	150°C	2.0	4.0
After exposure to processing solutions	A	4.0	4.0
B. 3/4-oz/ft^2 copper			
As received	A	4.0	4.0
After thermal stress	A	4.0	4.0
At elevated temperature	150°C	2.0	4.0
After exposure to processing solutions	A	4.0	4.0
C. 1-oz/ft^2 copper			
As received	A	5.0	8.0
After thermal stress	A	5.0	8.0
At elevated temperature	150°C	2.0	8.0
After exposure to processing solutions	A	5.0	8.0
D. 2-oz/ft^2 copper			
As received	A	6.0	10.0
After thermal stress	A	6.0	10.0
At elevated temperature	150°C	2.0	10.0
After exposure to processing solutions	A	6.0	10.0

See footnotes at end of table.

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TABLE I. Property values - Continued.

Property to be tested	Test condition 1/	Core thickness	
		Less than 0.030 in. to 0.010 in.	0.030 in. and over
E. 3-, 4-, and 5-oz/ft^2 copper As received After thermal stress At elevated temperature After exposure to processing solutions	 A A 150°C A	 2.0 2.0 5/	 12.0 12.0 12.0 12.0
2. Volume resistivity, minimum in megohm-centimeter After moisture resistance At elevated temperature	 F E-24/150	 10^6 10^5	 10^6 10^5
3. Surface resistivity, minimum megohm After moisture resistance At elevated temperature	 F E-24/150	 10^4 10^3	 10^4 10^3
4. Water absorption, maximum percent	E-1/105+des +D-24/23	0.010 - 0.20 0.020 - 0.20	0.030 - 0.20 0.060 - 0.10 0.090 - 0.08 0.120 - 0.08 0.240 - 0.05
5. Dielectric breakdown (parallel to laminations), minimum kV, step-by-step	D-48/50+ D-0.5/23	N/A	20
6. Permittivity at X band 2/ 3/	C-24/23/50	2.15 ±.04 4/ 2.20 ±.04 6/ 2.25 ±.04 2.30 ±.04 2.35 ±.04 2.40 ±.04	2.15 ±.04 4/ 2.20 ±.04 6/ 2.25 ±.04 2.30 ±.04 2.35 ±.04 2.40 ±.04
7. Loss tangent at X band 2/	C-24/23/50	0.0015	0.0015
8. Q (resonance), minimum	N/A	N/A	N/A
9. Flexural strength, minimum thousand lbs-per-square inch A. Lengthwise B. Crosswise C. Lengthwise at elevated temperature	 A A N/A	 N/A N/A N/A	 6.0 5.0 N/A

See footnotes at end of table.

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TABLE I: Property values - Continued.

Property to be tested	Test condition 1/	Core thickness	
		Less than 0.030 in. to 0.010 in.	0.030 in. and over
10. Arc resistance, minimum seconds	D-48/50+ D-0.5/23	120	180
11. Flammability, maximum burntime, seconds, each flame application	N/A	N/A	N/A
12. Dimensional stability maximum changes, inches/inch			
After stress	E4/105	$0.010\text{-}1.8X10^{-3}$	$0.030\text{-}1.5X10^{-3}$ $0.060\text{-}9.0X10^{-4}$
After bake	E2/150	$0.010\text{-}1.8X10^{-3}$	$0.030\text{-}1.5X10^{-3}$ $0.060\text{-}9.0X10^{-4}$

1/ See specimen conditioning requirements, conditioning tolerances and testing information table XXVI of MIL-P-13949.

2/ For the applicable test method see appendix I.

3/ Value shall be specified in ordering data.

4/ Tolerance of ±0.02 for all values of dielectric constant may be specified in the ordering data.

5/ Not practical for thin products due to flexing of sheet while conditioning.

6/ Materials other than 0.030 and 0.060 inch thick are not testable at 10 GHz.

7/ PTFE materials will not burn in small scale flame tests.

Length and width: Type GY shall be between 24 and 50 inches in length and between 14 and 50 inches in width.

Thickness: Cut sheets less than 24 inches by 14 inches shall meet the applicable thickness tolerances in 100 percent of the area of the sheet.

Marking: Internal marking not applicable.

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

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Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS, OS, SH
Air Force - 11, 85
NSA - S5
DLA - ES

User activities:
Army - AT, ME
Navy - MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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